ActInf Textbook Group ~ Cohort 8 ~ Session 8 (Chapter 4) ~ 4/10/2025

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Hello, welcome everyone to the active inference institute uh weekly textbook reading group. This is the latter session on Thursdays. Uh we are working with cohort 8 and we're looking at chapter 4 of the textbook. So as we know from chapter one our introduction, chapter two sort of the low road to active inference starting with uh basis theorem uh and kind of basian statistics or mechanics and then chapter three the high road to active inference uh drawing out from the imperative to minimize free energy in the free energy principle. Now we reach chapter four which is frankly a very kind of full chapter that attempts to in some ways tie together uh these two different like kind of strong like uh threads that we saw to active inference in chapters 2 and chapter 3 while simultaneously finally kind of introducing us to some core models that we see often in active inference. uh other aspects including various equations. We get to look at uh sort of graphical representations of models and so on. So I'll just try and give a quicker summary of this chapter. Um so with the intro we're introduced to the relationship between free energy and basian inference the form of typical active inference generative models and dynamics obtained from minimizing these models free energy. Specifically, we look at both discrete and continuous time models as well as the neurobiologically motivated idea of inferential variational message passing underlying the computations and uh the the general notion of predictive coding. So in chap uh section 4.2 two from basian inference to free energy. Uh recall that we are here working with approximate inference rather than exact basian optimality given the problem of intractability as well as neurobiological grounding of limited resources. So one can you know liken this to the notion of you know meta metabolism or you kind of like conservation of of of resources or minimizing expenditure in some ways in the sense of you know our brain is not fully connected. Uh it it is not a fully connected neural network. Uh and so there are various sparse connectivities going on functional connectivities and and different kinds of processes involved. And then from there just the general notion of us being able to take in every single bit of you know observation that comes into you know our visual space down to uh the idea of of of computing every single possible policy in a potentially infinite number of of worlds like you can start to imagine like you know that's not quite how this works. So that the variational methods are both mathematically useful as well as kind of a little bit more grounded in the the notion of like bounded rationality or bounded inference. We only have so many resources available to do the inference itself let alone maintain our homeostasis of which that would kind of be a piece. Um so from there we we're told that the the mathematics involved are drawn from uh general fields of linear algebra uh differentiation and and integration and calculus and the tailaylor series expansion in the case of continuous time models. And so the the textbook's appendices give further details on the relationships between these fields and computations. But chapter 4 here gives us kind of a broadstroke view of the maths involved. Um we know from equation 4.1 you know this is kind of the familiar basis theorem we've already been introduced to. Uh in this case we can kind of see how we could attempt to derive uh the probability of in this case the observations we receive u based upon sort of these this summation like summing all possible values of x and marginalizing uh such that we can start to kind of reach that value or or the other way around. Um and uh nonetheless we're immediately hit with this issue of intractability. And so that's where Jensen's inequality comes in. Um, this is, you know, in some ways can appear as sort of a clever trick, but at the same time, uh, if it's not quite a trick, if it's, uh, uh, not deceptive. And so what I mean by that is this is kind of what allows us to to formalize free energy as an a kind of lower uh, or excuse me, an upper bound on basian surprise. So in this case what we do is that instead of trying to directly compute um sort of model evidence we instead use this sort of variational posterior uh the the idiom is mind your P's and Q's uh in the sense that the Q is sort of this this variational posterior. Now this of course this term right here would resolve to one. It's more about what we're doing with it and how we're sort of operationalizing it in order to compute free energy or in this case negative free energy. Um, right? And so that's where this term is, right? That's that's what makes it an upper bound. The general notion is the log of an average is always greater than or equal to the average of a log. And it's because of that inequality that we're able to to to kind of formalize this in this way. So um see so we know that from see sorry there's just a lot of interconnectivity between the the equations in this in this textbook. So it's like equation 4.3 returning to basis theorem from before we're now rewriting it and taking its log. We can swap in free energy here. Yeah we already saw is a kind of tractable quantity. Um and then we can we're shown the idea of colac uh colac uh leler divergence which is kind of this core quantity that we're trying to minimize. It's kind of like if we can really minimize this divergence of our variational posterior from whatever our model says the the posterior kind of should be or however this was initially parameterized. then we can start to compute free energy. Well, not just compute it, but like if we can reduce this divergence, then we're effectively going to be also reducing variational free energy. So, um KL divergence. So, moving on from there, we're looking at various kinds of generative models. And in this case, I mean, so far we're not even necessarily in a kind of reinforcement learning paradigm. There's no sort of uh necessarily I mean you you could kind of swap some things around but ultimately like the this can be as simple you know they give us the example of like uh you know um a kind of a binary classification example for example um of of trying to kind of predict the the outcome does someone have you know a particular uh disease or illness based upon you know a test. So we can think of COVID, you know, and this was a one of the biggest drivers of uh sort of any kind of financial aspects of of creating tests and then and then for better or worse selling them on the on the market uh so that people could stay safe, but like how do you how do you devise um a a very solid like diagnostic COVID test? Well, you need to look at potential uh not just true positives and true negatives, meaning it's truly positive or truly negative whenever it says either of those in the test result, but also the other cases of a false positive or a false negative, right? So we can kind of in this case we might have like some kind of you know prior and then we we we take in an observation we end up computing you know we we can basically try and optimize these kinds of functions and and then in in terms of an actually existing COVID test. Um similarly we we would want to you know have it be essentially optimized in that sense like minimize all f false positives and negatives uh improve all the true positives and negatives and then if we start to think I I I kind of glossed over a lot of that but the the point is to to reach up to kind of the more uh exemplar active inference models and skip ahead just a little bit. So here we're looking at what in active inference we're always looking at priors right we're in a basium sort of framework and design. So whenever we see these components of like P parenthesis pi P uh parenthesis observations given states P the next state to plus one given or conditioned on the current state and the action taken. These all become kind of components of a model that are all kind of priors or you could also view them as models. There's an interesting question for anyone uh interested in kind of the um excuse me the equations and already at the beginning of this meeting there was a mention of like there was a kind of a discrepancy in the textbook textbook is uh unfortunately it's not perfect. I would say it's like as far as I can tell it's roughly 99% accurate, but there's a typo here and there. I believe there's a re-release of the textbook as well that addresses some of these things, but for anyone looking at the original one, unfortunately, there may be a a typo or two. And that's why we have things like the KOD and elsewhere to to see if we can uh answer some of these questions. And so um you know some people were were curious like what I just wanted to find this was interesting because of the the phrasing. Um what is pi? Page 69 the authors write at each time step the current state is conditionally dependent on the state at the previous time and on the policy pi currently being pursued. On page 71 they write thus we can interpret the priors equations 4.6 six combined with the likelihood of equation 4.5 is expressing a model pi of a behavioral sequence. So that's another tough aspect. Chapter 4 is a good chapter to really kind of get into in terms because now the the context here is that we're combining chapters two and three, right? We're trying to like we're kind of restabilizing after these two different paths that we took to to understand active inference. Now we're landing at kind of this central core of like well now we're talking about models and now we're talking about fuller action perception loops and and markoff blankets and so on. So in this case I mean as expressing a model this can be confusing. It's like well we had a generative model and then what is pi? Why why is pi this what appears to be an individual component in some way of one of these models? Why is that being called a model? Right? Is or or is it models all the way down? are these just model you know nested models and nested models and in some sense I mean if we can view you know these these components are so similar it's like you could almost extract these out of that larger model we were just looking at it it it is a bit of a perspective sort of thing because let's say prior over pi which we know from the terminology introduced to us in the textbook and I can also just tell you that pi typically represents pi policies. Policies being sequences of actions. Uh and so you could have multiple policies. One policy I have is to continue leading the the textbook and and going along with my notes here uh step by step. Another policy is to for whatever reason um you know take a break which which I won't be doing. But there are many policies I could do and all of them could have their own respective kind of trajectories uh beyond just the immediate next action but actually a whole sequence of actions that play together. Is this a model or is this you know a component of a model? It's it's kind of both because if you only looked at this alone, you you know, it could end up looking like a vector uh of different prob probabilities over different um actions or or it could be uh you know a matrix that has to do with the probabilities of a bunch of actions within different policies that then add up. So it is a model like this isn't a model of behavior. This is a probability of different policies. But once you kind of operationalize this in a broader context, it itself becomes part of a larger model. And so whenever we're putting together active inference models, we're thinking about things like like what what is your model? What what are its components? So I I'm focusing on this one a lot. This is a this we'll get to it again in chapter seven, but this is essentially a POMDP partially observable marov decision-making process. This three here uh this is your state transitions model, right? So it it is a model in in and of itself. It's a probability of your next hidden state given your current hidden state and policy. So it's like I do something I think something is going on at the next step. My three model here says that you know this will be the next hidden state the next thing that I believe this is how I think things transition in the world in relation to what's currently going on as well as my chosen action. It it it's describing what will come next. Right? So, so that's where the three comes in. You see it? So, we feed in the previous step uh state policy, we reach the next one. And then with two, this is where we have our observation model. And so, we have observations conditioned on states. What is the probability of my me noticing my stomach growling if I feel pretty certain that I am hungry? that I infer I'm hungry. What what is the probability if I think I'm hungry? What's the probability my stomach's going to growl? You know, um and so we put all these together and then we essentially run free energy minimization with this model and then it'll run free energy minimization to do two things. It'll um it'll infer states. So these are all being inferred. These aren't given to us. You know, there's maybe some true states in the environment, you know. So that maybe maybe this is trying to model, you know, I don't know if I'm hungry or not. I only have my bodily signals to let me know if I might be. Hunger is essentially a, you know, a very at the end of the day, it's a very kind of human linguistic concept based in real observations and and and occurrences in the world that we, you know, come into contact with. So whenever we minimize free energy, the notion here is that you know if this was a model of a person trying to always figure out if they're hungry or not based upon their bodily signals and then they decide once they are hungry oh they have a belief that if they're hungry they eat food then it will sate their hunger and so that you know this be satiation or being sated at the next time step it'll move that way and you can imagine infinitely more complex scenarios than that where we have many many different kinds of observations being fed in all at once you know and I was just giving a more discreet example of like it's like a binary hungry not hungry so from there we're also looking at um continuous time models and this is different right this would be this would be um you know you could measure someone's heart rate or or some other kind of continuous value over a whole slew of values or in some kind of continuous range um or or from negative infinity to positive infinity or otherwise and you do have various similar dynamics. The textbook uses different kind of nomenclature really just to remind you that this is the continuous case. So before we saw O's it's in the discrete state space scenario the POMDP or otherwise um O's are always for that Y's are always in the continuous space. So it's like you have your your O's versus your Y's, you have your S's versus your X's. And then we have V as you know and V is a kind of in a continuous state space. You can almost think of it as like a certain kind of level of like a force or something along those lines. like it could be a value that's taken you know we might have some prior to say like it is coming from a normal distribution or some other kind of uh distribution related to a continuous state space uh it's kind of just you know whenever you set up a model what do you think is the best kind of submodel almost that that that can act as priors in a basian scheme I have make a best guess that my force is Gaussian right and and you know it's going to be centered around zero on average and you know plus or minus one for the variance and you would do similar things for setting up the other priors. So anytime we do we create an active inference model we're always just kind of looking at these many subm models or components that themselves are prior and we set them up kind of as conjugate priors. meaning it for all intents and purposes for the sake of right now because I know I'm kind of going on for some time um you would want your your priors to be conjugate in the sense that conjugate is has a formal definition but to put it simply it's kind of like saying like it needs to be a good guess that makes sense and and so computationally it needs to sort of make sense. It's like if if my stomach is growling yes or no in this discrete space I'm not going to use you know a Gaussian for that the the value of.7 doesn't really align with me trying to say is my stomach growling or not or am I hungry or not uh maybe a Bernoli distribution for example would be a better case which is kind of like a distribution that describes a you know a p value a probability value and then for a for a binary. And so, you know, the probability of one of those options is P and the other one is one minus P, therefore Q. So, we we know that there we're working in more of a probabilistic space between two potential values. Uh, and then um we we we can kind of approach it that way. So, so Berni might make more sense. Um, not so much normal and maybe other way around for this or not or not. So those are the kinds of things we think about. Uh we already went through the POMDP with the nomenclature. It's you know it's it's we only see this nomenclature here and then it doesn't come back again until around chapter 7. But for anyone who may be interested um we saw here I mentioned earlier this was the likelihood model of observations given states. This can also be rewritten and this is supposed to be a shorthand. So it's supposed to be a useful uh kind of way of rewriting things. First cat, this is a categorical distribution or probability distribution in this case. Um it's over categories in a discrete state space, right? What what is the probability of my stomach is growling given I'm hungry? What is the probability that it's not growling given that I'm hungry? What is the probability that it's growling given I'm not hungry? And so we can go through the, you know, the four different possibilities there. And then those are kind of like four different categories. We we could assign a probability value to all four of those potential outcomes in this probability this likelihood model. And then it's frequently rewritten as the A matrix to just kind of simplify things. So you see, you know, if you wanted to talk about particular indices or particular things like uh I could be um uh stomach growling and J could be stomach not growling. So that's that's just kind of a note on the termin the the the sort of syntax here. We'll see similar things elsewhere. This is probability of a sequence of states condition on a policy and that is our B matrix our state transitions. This more clearly shows it but I think that they're trying to show like the kind of the mathematics behind it. It's a product and then other important components. D is one of the simplest. It's just kind of when you come to a situation where your priors over states. We're not even worried about observations. So O is not in here. We're not worried about policies. So pi is not in there. It's simply S. And they use the one to denote. It's kind of like this is your starting point because we always start somewhere. Uh there are debates in active inference about these kinds of things but but you know as a kind of neural process theory active inference doesn't necessarily give you you know here's the true magical starting point right we always come into a situation and free energy minimization plays out that's the general sense so if you're not sure what your starting point is you're going to have to pick some prior in some way and and so that's in a sense what this is could be an outcome of learning it over time. It could be quite similar to your last posterior inference, your last QS, um you know that then updated what your priors are now, but essentially it's just that um we would also see other important aspects. We haven't seen C yet. But just as D was the prior over uh states, we have C which is the prior over observations. And so that which also get relabeled as preferences. It's kind of like what you want uh so to speak. So I I I don't want my stomach uh to be growling. I do want it to not be growling. Um the reasons for that could be simply that that's kind of how my C matrix is constructed in some ways. This is this gets related to the notion of homeostasis. Uh in other ways you could view it as kind of like a set of attracting points because it's it's kind of like as we know from from other chapters if if we don't really have preferences if you don't have homeostatic step points that that you're attracted towards or if due to some kind of like suboptimal behavior behavior that's optimal to your given model but nonetheless behavior that is kind of suboptimal to the world around you and your relationship to it. you ultimately kind of succumb to entropic forces. I'm the one who cares about my hunger. Um the the trees outside don't so much care about that, right? And the the oxygen around me doesn't so much care about that. Uh it's just me. So, so I'm just kind of, you know, the whenever I'm trying to figure out my policies, my beliefs of what I should do, uh, a crucial component is going to be the C being in there so that I can judge, well, based on everything I'm seeing, how close is it to, you know, my preferences that relate to my homeostasis? is uh what's going on actually aligning with what I want to be occurring or uh what I hope to be occurring or this doesn't necessarily have to be described at all as a rational kind of conscious process in the sense of you know I have many auto autonomic processes occurring in my body that are doing things like uh kind of maintaining my blood uh oxygen flow and and regulation and and you know I'm not really particularly thinking about breathing right now and yet I'm doing it because I have various other processes in me. So, one way we could view that is these different sort of nested marov blankets, these different uh organ uh organs or or however you want to kind of slice it up. Poor use of words, but uh they're all minimizing free energy. That that would be the sort of framework here from active inference. They're all kind of doing their own thing. And then you know that from there we might say like maybe we hope that there's some kind of synchrony involved that that allows for uh almost as if we were looking at it like a multi- aent uh collaboration setting or something like that. Like we would hope that they're properly working together to where they can all kind of maintain their own respective uh homeostats and and and well-being so to speak, but also contributing to the broader system that makes up me as the the whole person or the whole whole body. Um who who is very, you know, dynamic in the world and always ever changing. Um so I think beyond that and we got to see G expected free energy was the the free energy of the future um which we were just looking at preferences are there pragmatic value crucial term information gain also crucial term two things to notice are first uh quite eerily similar to variational free energy But instead of running this over states to do state inference, oh my stomach is growling, I infer that my the the hidden state of my my body is that I'm hungry. Um instead, we're scoring for every available policy. It's expected free energy. Why is it expected? Because policies play out over time. And so long as we have some kind of state transitions model that adds this temporal dimension running left to right that says, "Oh, this is how I think things play out over time." And they even have a relationship to not just what the previous state was, but also what I do the policy. We see how that sorry, we see how that gets kind of insert that's inserted up here. So G is kind of trying to figure out which of these should happen based on my current inferences. It's how how is that going to happen? Well, we know that we want to minimize variational free energy from looking at the equations in chapter 3. Now we're at expected free energy and there aren't I mean you can look at certainly look at literature on free energy. There are other potential ways of computing it and different kinds of message passing schemes. But for the intents and purposes of this textbook which also gen this is the general paradigm is that there are two different kind of forms of free energy. Uh one for the moment moment by moment as it were variational free energy already saw in chapter 3. This is expected. Expected is going to be for policy inference rather than state inference. How do you def determine what to do? Another interesting thing is that both of these values are negative. This value is positive. We want to reduce this. So we actually want these values, we want information gain to go up. We're used to looking at uh variational free energy where you know complexity we want to reduce complexity and then accuracy we want to increase accuracy but it has a minus sign in front of it which is great because that means by flipping it around it's like we're reducing variational free energy we're reducing f by increasing accuracy meanwhile the complexity term is positive so we want to bring that uh down but here both of these are both of these are negative So, we actually want to bring up information gain. What is information gain? Uh-oh. This is uh Kobach Lebler divergence looks somewhat similar to the idea of reducing divergence. But here we're saying that we actually want to bring it up. Why would we want to increase divergence? This is the kind of exploration term that quote unquote naturally uh occurs or or balances uh uh exploration and exploitation in agent behavior in active inference agents. This is a crucial piece that kind of sets apart uh the behavior that we see in many other kind of standard reinforcement learning paradigms and models where oftentimes instead you're doing things like setting up some epsilon greedy term uh or or otherwise basically a value that says you know maybe 0.9% of point probability 0.9 you do what you think the best thing should be the best policy that should be 0.1 prob probability that you just do something random. And the concern there for reinforcement learning paradigms is oh well we can't just have a model who always jumps to what it thinks is the best thing. What happens if it you know arrives in some local minimum and then just stays there. It will never know how to get out of that suboptimal situation. It's optimal to the model. It thinks it's great. The model keeps going with what it thinks is best to do. But in the actual broader scheme in the environment that it's in uh you know it's it's actually not performing well or it's uh performing in a kind of suboptimal way right um so here this actually plays out naturally so to speak by saying oh what should I do well I want to figure out what's going on conditioned on what I saw and what I did versus what's going on versus what I did. So, this is kind of like saying, "Oh, I want to choose to do things that are actually very different from what I thought would was would be going on because I received these real observations from around me." It's kind of like saying I will choose to do thing go out of my way to do things to try and actually change my own beliefs about the the impact of my actions on the world around me. So imagine someone choosing to open up a textbook. It it is not getting you straight A's right now from just reading it. It's not it's it's not really accomplishing a lot in terms of keeping your homeostasis going, but in some ways it is certainly teaching you things and so that is information gained. So it's sort of a a simplistic way of looking at it but it's simply the sense of we you know in this sense this is a way of kind of quantifying mathematizing formalizing the sense of exploratory behavior. And just as with any other equation, this equation holds. So it's not this is not going to this equation itself is not going to change. So you don't have to manually define anything and you don't have to be creative by saying oh with 0.1 probability you just do something random and that's how I help you to not get stuck in a hole. Uh right and it can be broken up into to other terms too just as we saw with with variational free energy. So like expected ambiguity it's like what is the expectation or this posterior uh what is the expected entropy of a sequence of observations given a sequence of hidden state inferences. So it's it's kind of these kinds of terms like ambiguity it's kind of like it starts relating to the idea of one to many or many to many mappings. It's like, do if if if I'm going to go into a um a restaurant, do I expect to see that they're selling food or if I go into this building, do I expect it might be selling food or it might be selling tires or might be selling this or this? It's like I don't know if I want to go if I'm really hungry, I don't know if that would be the first building I would go into. I think I would go into the one where it seems very clear that they're selling like the food and then even better than that food that I want. Uh right, is another simple example. So um I think that we've touched on various things now you we've already touched on the categorical distributions uh we've touched on there's entropy term um we've touched on you know the the the significance of linear algebra um we have the softmax function which is frequently used softmax is quite it's often used in active inference to kind of um you we we we have to normalize our distribution so that they sum to one espe you know in the discrete state space setting. So if probability of you know being hungry is 0.5 then necessarily the the probability of not being hungry is also 0.5 you know 1 minus.5 is.5. Um here softax kind of uh normalized exponentials. So what we're doing is that we're kind of exponentiating uh these these differences. We're not just normalizing a distribution, but we're actually amplifying the values that are the strongest or or reducing the values that are the weakest or if there's already uh you know a lot of similarity between the values, it might actually flatten them. It's kind of like saying oh if you're if you're if if you're generally uncertain then it's kind of amplifying it to where you're you're very uncertain like almost like a uniform distribution and the other way around. So this this plays out in different ways. It plays out in expected uh not expected free energy. It plays out in policy inference as a way of sort of renormalizing everything. It's like okay you you have your your priors over your policies the E matrix. Uh you you have your um expected free energy over each policy. So into the future kind of related to deliberate planning and then you have variational free energy over each policy if you're including that in your equation. So then that's kind of moment by moment. So in policy inference it's like you have these uh these three different terms. Uh I don't I don't believe that particular yeah the particular equation is not oh okay this is it they've left out E. you don't necessarily have to include E in the sense of creating incilico you know kind of um experiments or or or uh simulations active inference uh kind of action perception loops but but more often than not you would include e here as well and it's kind of like saying what you do is in a sense a kind of uh normalized exponentiated function of the policies uh that you scored based on expected free energy minim minimization. G uh the way you scored them based on variational free energy uh you know complexity minus accuracy. Um and then E would be your habits, your priors over policies. They write it in this way I believe to just kind of like make a point. This is like an introductory sort of point over um sort of how F and G both can play into what you choose to do. But I've always been fascinated by the fuller write out of this uh equation from sort of like a behavioral or social science perspective because what it's saying is that if you did have the e here, it's like saying what you do is in some way a kind of balanced uh uh comp uh a kind of balanced um outcome of what you deliberately plan to do versus f. So just maintaining yourself in the moment. Uh keeping things like as simple as possible and no simpler while still being accurate. Uh and then also your habits. So what what what you've done in the past, what's going on in the moment, and also what you plan to do for the future, right? Which goes a little bit further than uh sort of the the conaman thinking fast and slow scheme. I I saw I was uh whenever I ran a workshop last year, there was a professor who was asking about the relationship there. And I'm like, well actually we have we have three different component. We we kind of have the past and the present and the future all kind of here in their own respective ways instead of just the the the diatic relationship between a fast and slow system. We start getting into variational message passing. So we'll in belief propagation these things come up much more strongly next chapter which is why I didn't u put too much emphasis on those for right now. But we get a little bit of a a little bit of um a preview suppose the the notion would be that we're just passing. So we have different prediction errors and uh hidden state inferences uh at different time steps one to another over the course of different observations that are received sequentially. states that are inferred time. And that's general sense. We're shown generalized coordinates of motion. Those will come up much more in chapter 8 on continuous time models. It's like, well, how do we figure out state transitions if we're not just going from hungry to not hungry to hungry to not hungry? Instead, it's a whole range of values potentially. And it's we're essentially we're just looking at derivatives which those who've done you know differential equations and calculus and in in other settings uh including you know working with neuroiming data or otherwise this will probably not come off as particularly foreign uh to them. But the the notion would be that we'd add some number of Taylor series expansion kind of terms. uh more often than not they claim that around six are sufficient all depending on the way that you're fitting your model and what seems to hold. So I'll kind of stop there. We do we do see some more relationships between you know other fields other other ways of doing things. we have uh maximum a posterior uh estimates for for states and and relating it to that um lelass approximations and yeah so it it goes on for some time but it's it's it's just there's so much of this chapter that they try to fit in all at once while knowing that the next chapter is going to get into message passing the chapter after that is going to get into the bare bones of modeling chapter Chapter seven is going to get into the POMDP model. Chapter eight's going to get into these continuous time models. Chapter nine will get into kind of the fun uh aspects of of of data analysis and what you can actually do from with the results of these models. So, I'm going to I'm going to leave it there. Apologies for anyone if if they um had anything uh they really wanted to get into that I might have spoken over. But I'll I'll go ahead and open up the floor if anyone has any uh talking points or anything interesting that stuck out to them whenever they're reading this textbook and maybe we can have a look at it. Thanks. So the the joint probability distribution is the thing generated by the generative model, right? Yeah. So the joint distribution between states and observations, we might be able to pull up a representation of that, but essentially like a the probability of X comma Y or the probability of S comma O um is essentially your generative model, right? It's a it's a model over states and observations together. And it's the way that we sort of optimize it and figure out, you know, the posterior states if you're trying to infer states, which is sort of the the active inference way, but also the basian way in general, is to kind of break apart that joint distribution and apply different forms of marginalization and variational inference in order to actually use it to to compute posteriors. it to make sure I really answer your question though just yes the the generative model the agent the organism in question is is essentially a joint distribution over over states and observations it is of course a little more complex because in the the textbook you know the the way that they show basis theorem is that we're repeatedly relating states and observations but really once we bring in the capacity for planning and decision making and taking actions where suddenly it's like well pi where was that in basis theorem it's like well it's still a basian framework it's still we still have a prior for it and whenever it comes to policy inference we still infer policies but we're using expected free energy to as as kind of the primary means of doing that um all this is to say you could write a generative model very differently uh that had more components. So it could be a joint distribution of um I liked this if I could find it. And for anyone interested in programming um yeah anyone who who's interested in Python programming or otherwise or wants to take a crack at discrete state spaces um this has been uh for two to three years um a uh kind of the the primary library if there were a primary library for uh doing active inference in Python but only for discrete state spaces. So unfortunately you're not going to be able to get into your your Gaussian distributions necessarily but um here they write a generative model I've sort of liked this nomenclature um with a a slight discrepancy that I want to mention. So this writing of it you we're we're familiar with the fact that this is a prob a probability distribution. It's wrapped with a capital P. We have O for observations, S for states. As before, we have U, which that's, you know, uh that's something to watch out for in active inference literature. Sometimes people decide to use sort of their own uh terms. Th this one kind of makes sense because sometimes the textbook will use the letter U to denote an individual action. So a se a sequence of use or a sequence of actions then denotes a particular policy with a policy being a sequence of actions. So this this from the textbook's perspective this would be taken as an action but here they're kind of using it as the more general this is a policy and then this u this final kind so this fee is for hyperparameters what we haven't even talked about hyperparameters yet in at this point in the in the textbook but the idea is you could kind of say oh here are all the parameters specific to this model not just its priors or or its understanding or or the way that it processes observations and states and policies, but also what are all these little specific aspects, you know, uh it's it's uh again not in chapter 4, it's in the second half of the book, I think, but the we I believe we've already been introduced at least to the concept of learning as opposed to inference. With inference, we've already talked about it a lot. We're inferring states from minimizing variational free energy f. We've talked about minimizing um expected free energy to figure out what to do in policy inference. Uh learning however would be what happens whenever one of your priors or some kind of parameter on it itself changes. So that could be precision values for example. Precision values what are those? They are neurobiologically generally an attempt at trying to model sort of neurotransmitters and transmission or synaptic gain uh in different ways um based on you know some of the empirical work that Fristen Schwenbeck and others have done. Uh one way that you can view expected free energy as a form of deliberate planning is that whenever it's uh kind of modulated by this uh they use the term gamma but it is a a precision value. And so if it goes up it's kind of relatable to sort of dopamine uh in the sense that dopamine tends to kind of with with a depletion or or diminishment or or or or insufficient dopamine so to speak or maybe the the re-uptake process is not quite strong enough. What happens is that you end up with a person who's very lethargic, right? And so there's not a lot of confidence. They might have ideas of what they should do, but they won't necessarily feel confident in their capacity to, you know, fulfill them, to get up and do what it is they need to do. You know, in in very extreme levels, it starts to more or less mirror, you know, the kinds of behaviors we see in like major depressive disorder from a psychiatry perspective. Um and and then on the other hand, a very extreme amount of dopamine would make someone very very enthusiastic about the policies that they kind of, you know, they they would be almost excessively confident in what they're going to do. And that can kind of manifest manifest in v uh various ways as well including like leading to different forms of delusions uh or or even like kind of um kind of psychosis or or different kinds of hyperprecise priors that are they're so specific and and are difficult to sort of unlearn or modulate appropriately that it leads to what we could kind of definitively defined as pathologic behaviors. I don't mean to scare everyone by giving those extreme examples. Um but the but the notion is that precisions precision values would be something that kind of modulate the strength or the certainty or the confidence of each of these kind of inter nested models your policy or behavioral model your likelihood model or your state transitions model or what other other component you have. Um and then this itself could change. You could put a prior on this prior. um in which case you would in a discrete state space setting you'd use a matrix derish distribution which is kind of like a categorical distribution but it doesn't have to be normalized. So it you it could start to kind of collect observ it's kind of like saying yes this is a marov environment like a kind of marov marovian paradigm we're looking at here going from you know from one step to the next and the next step here doesn't necessarily fully depend on what came before but nonetheless you can kind of build up observations along the way and build them up in your hyperparameter what does that mean in more plain terms it's kind of like saying we can still follow this paradigm And yet long-term memory exists um and and it matters but the way that it matters is that it changes how the model works from time step to time step or at least as frequently as the the model is learning. So um so essentially learning would involve actually changing the parameters of the model itself. So fe as a kind of standard this is a fuller representation of what a generative model might look like in in how we often do active inference. You have observation states but you also have essentially policies or behav the behavioral component and then you have all these different hyperparameters you may or may not have included in constructing your your model. And so like with with PIMDP what would you do there? This itself can be kind of ex helpful or explanatory. Um but the idea is that you're going to compose an agent. It's going the agent can infer its states can infer states. Infer S or infer X. Um here we'll say S for the discrete state space setting. How does it do it? Well, it takes in a new observation. So it takes in an observation right here. And we're going to call this function infer states. Infer states is incredibly complex. Um, and so I've been working with a PhD student recently and and and he and I were very excited to work with PMDP, but then we noticed we might need to make some modifications and it's like, well, actually in first states it's quite complex underneath. So that's been a little bit of a a journey, but it's been it's been fun and pretty enlightening. Um, there's always more to learn. So you would set up an agent by supplying it with its A matrix and these other matrices I mentioned earlier. The A the likelihood matrix like probability of observations given states the B matrix which would be the probability of the next state given the current state and the thing I chose to do policy. Uh C. So you're telling your you're saying what your agent's preferences are over observations it receives priors over observations. And then D prior over states where where where it's starting point about what it believes is currently going on might be uniform might be completely uncertain or it might be very certain that something is already going on and then it'll be yet to be seen how it responds to receiving a new observation and then inferring states. So um uh so I had another question. Uh so you you you have these you have these probabilities that are part of your model, but how do you know that the model isn't just like I don't know an AI hallucination or something? Like how how do you know that the underlying states of your model are any good? Um, how do you know if the like is it kind of like how do you know if your model is uh so somewhere else in the book it says there's a difference between the states of your model and the states of your actual world? How do you know they're not like really different? That's the tough part. Uh you might not. And what I mean by that is um in the in the same sense that you know because and I apologize I don't mean to sound strange but I'm just kind of using metaphors here uh or or analogies. It's kind of like if you have a you know the existence of science as a field involves people going out gathering observations to infer hidden states what they think about the world and and they can do it repeatedly and they they consider that to be evidence that they're pulling in for you know their theory. So you can view that as an agent a scientist here is an agent. They're collecting observations to infer hidden states of the world if they want to understand uh climate, you know, and they use different kinds of scientific implements to gather that as evidence in order to make an inference about kind of, you know, would you know, make a forecast about what's going to happen or what, you know, or me pulling out my phone to look at what the temperature is right now to make a best guess about what the temperature is outside. This is not in some ways dramatically different than that that you know and this is an agent who takes an observation and infers states. So, so to say something in here might be bad, it's like well um you would kind of need some you might need some external observer for example who who does know kind of a closer a better approximation to the true state of the environment and they could tell you if it's it's bad or not. But um you're ultimately it doesn't matter if you're using active inference or if you're using a deep learning reinforcement learning u model or agent. It's like it's the it's kind of the same thing either way. You just kind of you set up a model that you think is most appropriate to the situation based on previous research which itself is an observation for you to infer your own states about what the priors on this thing should be. And it's kind of like the you see that that's the kind of nestedness that we get out of looking at things from the active inference sort of framework. Everything we take in is an observation. and we just infer states. And so, um, I mean, or you could look at it in the context of a particular simulation. So, you could say, oh, there's there's an agent and we want them to, you know, uh, oh, we have a we have a a mouse and a teammates, right? And it despite the fact that it keeps going left and it keeps missing the cheese and it keeps instead get getting a a negative stimulus like a little shock, it still keeps going left. Why is that? For us, the observers, we think, well, that's definitely suboptimal behavior that the rat wants the cheese. Why does it keep going left even though it's getting contradictory evidence repeatedly that the cheese is in fact not on the left? In that case, we might make a a a kind of more conclusive statement that this is suboptimal. And then we could from there say like, oh, if we view the mouse as a generative model or at least being equipped with a generative model that it's using to try and solve that problem of getting the cheese or navigating the maze, then in that case you could say like, oh, that that's bad. Like there are bad priors in this model. Like that might be one for way of describing like how you might have criteria for that. Um if you want a model that you're in in a machine learning paradigm, you often take you know you take data and you split it into a training set and a testing set and then you train the model in the training data and then you test it by looking at how well it performs versus the true values in the testing data, right? And you can use some kind of metric like mean root mean squared error or or otherwise depending on your this the states basically whatever error metric is most appropriate for that scenario and then that would be a kind of measurement criteria to say how good or bad your model is right and then from there it's like well for a particular problem like you know trying to predict the MNIST data set it's like well maybe there's some models that are almost perfect at it and others that aren't. But then maybe there's another problem that's very very hard to where the best model that anyone has ever published on for that particular problem at hand. They don't the best there a model that they could do was a model that's about 50% accurate which would be a coin flip which would not be useful um in a sense but uh but it's it just kind of can go on and on. So it's it's more about getting used to the the idea of thinking about modeling what it is we're trying to do whenever how often can you find would you say uh a model like this for that is that is significantly not just better than a coin flip but like significantly better by by some criterion uh to model something complex like like a cat how how do you get a model it's like better than I don't know how do you make a model that can that's good enough to predict it for a while. Yeah, that's a good I mean it's a good question and it's honestly it's a very broad I mean anyone who does modeling it it's there are many different kinds of answers to that question. Um ultimately we all kind of want to achieve that which is why I keep speaking so openly instead of pointing at formulas. Um I mean if one way to do it from the active inference perspective it's like well you can just you know figure out what you think the cat if you're trying to model a cat or a cat's behavior right then what you could do is you could say well here are some things I think the cat would do or policies available to the cat maybe it runs maybe it scratches maybe it you know such and such or you could look at a continuous state space setting where it's like oh it has multip multiple things it does each with their respective kind of continuous values like how hard does it scratch or how fast does it run when it runs. Um which can vary, right? And that's kind of the utility of using the continuous state space. Um, but you just kind of go with your best guesses of how you think that would work first. And then second, you would, you know, in in reality, if you're trying to model some kind of realw world phenomena, then you would want to find some kind of data from the cat that you can actually record and and use sort of as your data that you will use to fit your model to it. Right? The notion here in active inference is that we're not trying to model an entire brain or the entirety of an organism. That would kind of start to defeat the purpose of some of the variational methods going on here and some of the the the the I mean you could theoretically, but but the idea would be that we're trying to model the the the we're trying to model the model that the agent uses to solve a problem. either maybe it's a cat in an open world where there isn't really a very specific problem. But if you view it as well the cat also has to breathe oxygen and it also has to eat food and it also has to sleep you know other things that it needs to do in order to stay alive to maintain its homeostasis. Then you already start you start to have the trappings of a C matrix like that that those would be the homeostats the preferences and and then you can think of what it would do to accomplish and then from there if you fit the model to real data from Oh yeah sure sure I'll go ahead and stop John we can continue this conversation some other time I appreciate your questions um thanks everyone more questions on Discord Thanks all. See you. Thank you, Andrew. Bye.